

Dr. Doc:

 Welcome to JU pathology Exam 

#2014 Pathology Exam

1. Which of the following statement is true?

- A. Infection has no role in chronic bronchitis
- B. Asthma is usually fatal
- C. Smoking is common risk factor for mesothelioma
- D. Bronchiectasis more commonly affects the upper lobe of the lung
- E. B and D are correct
- F. None

Answer: F. None

Explanation:

All of the statements A–D are incorrect.

▼ A is incorrect:

 Infection plays an important role in chronic bronchitis, particularly in maintaining chronic airway inflammation and producing acute exacerbations.

▼ Cigarette smoking is the major underlying risk factor, but respiratory infections frequently worsen the disease.

▼ B is incorrect:

👉 Asthma is generally a chronic but manageable disease.

👉 Although severe asthma attacks, particularly status asthmaticus, can be fatal, most patients do not die from asthma.

▼ C is incorrect:

▼ Smoking is a major risk factor for bronchogenic carcinoma, but it is not the principal risk factor for malignant mesothelioma.

▼ The major etiologic association with mesothelioma is occupational asbestos exposure.

▼ D is incorrect: Bronchiectasis commonly involves the lower lobes, although the distribution can vary depending on the underlying cause.

▼ Upper-lobe involvement may occur in conditions such as cystic fibrosis.

★ High-yield points:

🎯 Asbestos → malignant mesothelioma

🎯 Smoking → COPD and bronchogenic carcinoma

🎯 Bronchiectasis → permanent bronchial dilation, commonly involving lower lobes

🎯 Asthma → usually reversible; severe status asthmaticus can be fatal.

▼ Lung pathologies have specific anatomical predilections e.g.

👉 centriacinar emphysema in upper lobes ▲

👉 bronchiectasis and panacinar emphysema in lower lobes ▼

2. From first to last, the morphological changes in lobar pneumonia occur in which correct chronological order?

- A. Grey hepatisation, red hepatisation, congestion, resolution
- B. Red hepatisation, grey hepatisation, congestion, resolution
- C. Congestion, grey hepatisation, red hepatisation, resolution
- D. Congestion, red hepatisation, grey hepatisation, resolution

Answer: D.

▼ Congestion → Red hepatisation → Grey hepatisation → Resolution

Explanation:

▼ Lobar pneumonia classically progresses through four morphological stages.

1. Congestion:

▼ This is the earliest stage.

▼ The affected lung  becomes heavy , red , and boggy  because of vascular engorgement and accumulation of intra-alveolar fluid containing relatively few neutrophils.

2. Red hepatisation:

▲ The alveoli become filled with red blood cells, neutrophils, and fibrin.

● The affected lung becomes firm and liver-like, which explains the term "hepatisation."

3. Grey hepatisation:

▲ Red blood cells begin to break down while fibrin persists.

○● The lung becomes grey-brown and remains firm because of the continuing fibrinopurulent exudate.

4. Resolution:

○ Enzymatic digestion breaks down the intra-alveolar exudate.

▼ The material is removed by macrophages, coughing, and lymphatic drainage, allowing restoration of normal lung architecture.

★ High-yield points:

✦ Sequence:

Congestion



● Red hepatisation



○ Grey hepatisation



Resolution ✓

● Red hepatisation = RBCs + neutrophils + fibrin

○ Grey hepatisation = RBC breakdown + persistent fibrinopurulent exudate

Lobar pneumonia affects a large portion or an entire lobe uniformly, unlike the patchy distribution of bronchopneumonia.

✓ Most common outcome = resolution

3. Which disorder is characterized by permanent dilation of bronchi and bronchioles?

A. Chronic bronchitis

- B. Emphysema
- C. Bronchiectasis
- D. Atelectasis

Answer: C. Bronchiectasis

Explanation:

▼ Bronchiectasis is characterized by permanent and irreversible dilation of bronchi and bronchioles resulting from destruction of the muscular and elastic components of the bronchial wall.

🔥 The underlying process is usually chronic inflammation and infection, often associated with impaired clearance of respiratory secretions.

✳️ Persistent infection damages the bronchial wall, producing progressive dilation and accumulation of purulent secretions.

⚡ Important causes include cystic fibrosis, bronchial obstruction, recurrent pulmonary infections, and congenital disorders affecting mucociliary clearance.

⚡ Clinically, patients commonly present with chronic cough and copious, purulent, sometimes foul-smelling sputum.

★ High-yield points:

✳️ Bronchiectasis = permanent bronchial dilation

⚡ Major mechanism = destruction of bronchial wall by chronic infection/inflammation

⚡ Classic presentation = chronic cough + copious purulent sputum

✳ Cystic fibrosis is an important cause

👤 Bronchiectasis is defined as the permanent dilation of bronchi and bronchioles caused by the destruction of muscle and elastic supporting tissue.

▼ Bronchiectasis is usually secondary to chronic necrotizing infections or persistent obstruction.

4. A patient receives cardiopulmonary resuscitation and suffers aspiration of gastric contents. Over the next 10 days he develops a non-productive cough along with a fever to 37.9°C. A chest radiograph reveals a 4 cm diameter mass with an air-fluid level in the right lung. A culture done for the sputum shows mixed flora. Which of the following conditions is he most likely to have?

A. Squamous cell carcinoma

B. Lung abscess

C. Chronic bronchitis

D. Bronchiectasis

Answer: B. Lung abscess

Explanation:

👉 The patient has several classic features of an aspiration-related lung abscess.

⚡ Aspiration of gastric or oropharyngeal contents can introduce bacteria into the lower respiratory tract. The resulting infection causes localized suppurative destruction of lung tissue, producing a cavity filled with pus.

🦋 The chest radiograph showing a cavity containing an air-fluid level is highly characteristic of a lung abscess. The mixed bacterial flora also supports aspiration because aspiration-related abscesses are frequently caused by mixed aerobic and anaerobic organisms.

🦋 The right lung is commonly affected because the right main bronchus is wider, shorter, and more vertically oriented than the left main bronchus.

★ High-yield points:

▼ Aspiration → common cause of lung abscess

▼ Lung abscess → localized suppurative necrosis

▼ Air-fluid level → classic radiologic finding

Aspiration abscess → often mixed bacterial/anaerobic flora

▼ Right lung is commonly affected

5. A 30-year-old male with cystic fibrosis presents with shortness of breath and production of a large amount of foul-smelling sputum. What is the most likely cause?

A. Atelectasis

B. Asthma

C. Bronchiectasis

D. Adenocarcinoma

Answer: C. Bronchiectasis

Explanation:

▼ Cystic fibrosis causes abnormal production of thick, dehydrated mucus because of defective CFTR-mediated chloride transport.

▼ The thick secretions impair mucociliary clearance and predispose the respiratory tract to recurrent bacterial infections.

▼ Repeated infection and inflammation progressively destroy the muscular and elastic components of the bronchial walls, resulting in bronchiectasis.

▼ Bronchiectasis produces dilated airways that retain large amounts of infected secretions.

▼ Therefore, patients commonly develop chronic productive cough with copious purulent or foul-smelling sputum, recurrent infections, dyspnea, and sometimes hemoptysis.

★ High-yield points:

🎯 Cystic fibrosis → thick mucus → impaired mucociliary clearance → recurrent infection → bronchiectasis

Bronchiectasis → copious purulent/foul-smelling sputum

▼ *Pseudomonas aeruginosa* is an important pathogen in CF.

6. The most common outcome of treated bacterial pneumonia is?

- A. Resolution
- B. Scar formation
- C. Abscess formation
- D. Bronchopleural fistula

Answer: A. Resolution

Explanation:

▼ The most common outcome of appropriately treated bacterial pneumonia is complete resolution.

🧬 During resolution, inflammatory exudate within the alveolar spaces is enzymatically digested.

▼ Macrophages remove cellular debris, and the remaining material is either expectorated or drained through lymphatic pathways.

▼ Because the basic alveolar framework is usually preserved, the lung can return to normal architecture without significant scarring.

▼ Complications such as abscess formation, empyema, organization, bacteremia, and bronchopleural fistula can occur but are much less common.

★ High-yield points:

✦ Most common outcome of bacterial pneumonia = resolution

✦ Pneumonia usually heals without significant fibrosis

Complications include abscess, empyema, bacteremia, and organization

7. A 60-year-old male with a history of smoking presents with hoarseness of voice. Laryngoscopy reveals a 3 cm laryngeal mass. On histopathologic examination, the mass was identified as malignant neoplasm. The most likely histologic diagnosis is?

A. Adenocarcinoma

B. Small cell carcinoma

C. Squamous cell carcinoma

D. Lymphoma

Answer: C. Squamous cell carcinoma

Explanation:

▼ The overwhelming majority of laryngeal malignant tumors are squamous cell carcinomas.

▼ The major risk factors are cigarette smoking and alcohol consumption, with the risk increasing substantially when both are present.

✦ Laryngeal SCC commonly occurs in older men.

🐎 Hoarseness is particularly important because tumors involving the true vocal cords interfere with vocal cord vibration.

▼ Persistent unexplained hoarseness in an older smoker should therefore raise suspicion for laryngeal carcinoma.

★ High-yield points:

▼ Most common laryngeal malignancy = squamous cell carcinoma

🎯 Major risk factors = smoking 🚬 + alcohol 🍷

▼ Hoarseness = classic early symptom, especially with vocal cord involvement

Persistent hoarseness in a smoker requires evaluation

8. The most common causative organism for community-acquired typical pneumonia is?

A. Streptococci

B. Pneumococci

C. Haemophilus

D. Staphylococci

Answer: B. Pneumococci

Explanation:

▼ *Streptococcus pneumoniae* (pneumococcus) is the classic and most common cause of community-acquired typical bacterial pneumonia.

Pneumococcal pneumonia classically produces an acute illness with fever, productive cough, pleuritic chest pain, and radiographic consolidation.

✓Morphologically, it can produce the classical stages of lobar pneumonia: congestion, red hepatisation, grey hepatisation, and resolution.

★ High-yield points:

▼ Most common classic cause of community-acquired typical pneumonia = *Streptococcus pneumoniae*

🏠 Typical bacterial pneumonia → alveolar consolidation

Pneumococcus → classically associated with lobar pneumonia

9. Emphysema associated with alpha-1 antitrypsin deficiency is?

A. Panacinar

B. Distal acinar

C. Centriacinar

D. Irregular

Answer: A. Panacinar

Explanation:

▼ Panacinar emphysema is strongly associated with alpha-1 antitrypsin deficiency.

✦ Alpha-1 antitrypsin is a protease inhibitor 🚫 that normally protects lung tissue from enzymes such as neutrophil elastase.

➡ Deficiency allows excessive proteolytic destruction of alveolar walls.

● Panacinar emphysema involves the entire acinus relatively uniformly, from the respiratory bronchiole through the alveolar ducts and alveoli.

☑ It has a characteristic predominance in the lower lobes, especially in severe alpha-1 antitrypsin deficiency.

☑ In contrast, centriacinar emphysema is strongly associated with cigarette smoking and predominantly affects the upper lobes.

★ High-yield points:

🚫 Alpha-1 antitrypsin deficiency → panacinar emphysema

Panacinar → lower lobes

🚬 Smoking → centriacinar emphysema → upper lobes

➡ Alpha-1 antitrypsin normally inhibits neutrophil elastase.

11. The most common etiologic factor implicated in chronic bronchiolitis is?

- A. Air pollutant
- B. Mycoplasma infection
- C. Cigarette smoking
- D. Cotton mills

Answer: C. Cigarette smoking

Explanation:

🚬 Cigarette smoking is the most important etiologic factor associated with chronic bronchitis and chronic small-airway disease, including chronic bronchiolitis.

▼ Repeated exposure to tobacco smoke causes chronic inflammation of the bronchi and bronchioles, increased mucus production, mucus plugging, and structural changes in the airway walls.

➡ These changes increase airflow resistance and contribute to obstructive lung disease.

★ High-yield points:

🚬 Cigarette smoking = major cause of chronic bronchitis and chronic bronchiolitis

▼ Chronic bronchiolitis → inflammation + mucus plugging + airway narrowing

✳️ Smoking is the major preventable risk factor for COPD.

10. Asbestos exposure results in all of the following changes except?

- A. Interstitial pneumonia
- B. Malignant mesothelioma
- C. Pleural effusion
- D. Pleural fibrosis

Answer: A. Interstitial pneumonia

Explanation:

▼ Asbestos exposure causes several characteristic pulmonary and pleural abnormalities, including asbestosis, pleural plaques, pleural fibrosis, pleural effusions, and malignant mesothelioma.

✓ The classic parenchymal lesion is interstitial pulmonary fibrosis, known as asbestosis.

💡 Therefore, "interstitial pneumonia" is not the appropriate characteristic asbestos-related lesion in this context.

▼ Asbestos fibers become coated with iron-containing material and proteins, producing characteristic asbestos bodies, which appear as golden-brown ●, beaded or dumbbell-shaped structures microscopically.

★ High-yield points:

○ Asbestos → interstitial pulmonary fibrosis (asbestosis)

○ Asbestos → malignant pleural mesothelioma

○ Asbestos → pleural plaques and fibrosis

● Asbestos bodies = golden-brown coated fibers

▲ Smoking + asbestos → markedly increased risk of bronchogenic carcinoma

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Explanation:

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+ Repeated exposure to tobacco smoke causes chronic inflammation of the bronchi and bronchioles, increased mucus production, mucus plugging, and structural changes in the airway walls.

↗These changes increase airflow resistance and contribute to obstructive lung disease.

★ High-yield points:

▼ Cigarette smoking = major cause of chronic bronchitis and chronic bronchiolitis

✦ Chronic bronchiolitis → inflammation + mucus plugging + airway narrowing

Smoking is the major preventable risk factor for COPD

12. Among the following genes, which one is mostly affected in small cell lung carcinoma?

A. RB gene

B. CDK gene

C. MYC gene

D. RAS gene

Answer: A. RB gene

Explanation:

▼ Small cell lung carcinoma (SCLC) is strongly associated with inactivation of the RB1 and TP53 tumor suppressor genes.

▼ Loss of RB function removes an important checkpoint controlling progression through the cell cycle.

🧬 Loss of p53 further compromises DNA-damage responses and apoptosis.

✦ These genetic abnormalities contribute to the highly aggressive behavior of SCLC.

🎯 SCLC is strongly associated with cigarette smoking, has a high mitotic rate, extensive necrosis, and tends to metastasize early.

★ High-yield points:

✦ SCLC → RB1 and TP53 abnormalities

✦ SCLC → strongly associated with smoking

✦ SCLC → aggressive tumor with early metastasis

✦ Azzopardi effect → classically associated with SCLC

13. Which of the following organisms is a common cause of pneumonia in patients with cystic fibrosis?

- A. Haemophilus influenzae
- B. Pseudomonas aeruginosa
- C. Legionella pneumophila
- D. Moraxella catarrhalis

Answer: B. Pseudomonas aeruginosa

Explanation:

▼ Patients with cystic fibrosis have thick, viscous respiratory secretions because of abnormal CFTR function.

▼ These secretions impair mucociliary clearance and create an environment favorable for persistent bacterial colonization.

🦠 *Pseudomonas aeruginosa* is particularly important in patients with established cystic fibrosis.

💡 Chronic colonization can cause recurrent pulmonary infections and progressive structural lung damage, especially bronchiectasis.

★ High-yield points:

✚ Cystic fibrosis → *Pseudomonas aeruginosa* is a major pulmonary pathogen

✚ CF → impaired mucociliary clearance

🎯 Chronic *Pseudomonas* infection → recurrent pneumonia + bronchiectasis

14. Restrictive lung diseases are characterized by all of the following except?

A. Shortness of breath

B. Cyanosis

C. Tachypnea

D. Wheezing

Answer: D. Wheezing

Explanation:

✦ Restrictive lung diseases are characterized primarily by reduced lung volumes, particularly reduced total lung capacity.

✦ The lungs are difficult to expand because of either intrinsic parenchymal disease or extrapulmonary causes.

✦ Patients commonly present with dyspnea and tachypnea, and severe disease can result in hypoxemia and cyanosis.

✦ Wheezing, however, is more characteristic of obstructive airway disease, where narrowing of the airways produces turbulent airflow.

Examples include asthma and COPD.

🎯 In pulmonary function testing, restrictive disease typically shows reduced FVC and TLC with a normal or increased FEV1/FVC ratio.

★ High-yield points:

✓ Restrictive disease → ↓ TLC

FEV1/FVC → normal or increased

✳ Obstructive disease → ↓ FEV1/FVC

👉 Wheezing → more characteristic of obstructive disease

15. Lung cancer that produces hypercalcemia is mostly?

- A. Squamous cell carcinoma
- B. Adenocarcinoma
- C. Small cell carcinoma
- D. Carcinoid tumor

Answer: A. Squamous cell carcinoma

Explanation:

👉 Squamous cell carcinoma of the lung is classically associated with hypercalcemia as a paraneoplastic syndrome.

▼ The mechanism is usually production of parathyroid hormone-related peptide (PTHrP) by tumor cells.

🧠 PTHrP increases calcium levels by promoting bone resorption and increasing renal calcium retention.

▼ This is an important distinction among lung cancers. Squamous cell carcinoma is associated with PTHrP-mediated hypercalcemia, whereas small cell carcinoma is more strongly associated with SIADH and ectopic ACTH production.

★ High-yield points:

▼ Squamous cell carcinoma → PTHrP → hypercalcemia

▼ SCLC → SIADH and ectopic ACTH

Lung cancer can produce important paraneoplastic syndromes

16. The most common cause of death in Goodpasture syndrome is?

A. Respiratory failure

B. Uremia

C. Hemoptysis

D. None

Answer: B. Uremia

Explanation:

▼ Goodpasture syndrome is an autoimmune disease caused by antibodies directed against the glomerular and alveolar basement membranes.

▼ These antibodies recognize the alpha-3 chain of type IV collagen.

🔑 The disease produces a combination of rapidly progressive glomerulonephritis and pulmonary hemorrhage.

👉 Although pulmonary manifestations such as hemoptysis can be dramatic, progressive renal injury can result in severe renal failure and uremia, which is a major cause of death if untreated.

★ High-yield points:

✓ Goodpasture syndrome → anti-GBM antibodies

Target → type IV collagen in glomerular and alveolar basement membranes

🏥 Clinical combination → pulmonary hemorrhage + rapidly progressive glomerulonephritis

✚ Renal failure → major cause of mortality

17. Which of the following is not a typical feature of asthma?

- A. Reversibility
- B. Asthma is recurrent
- C. The severity of asthma increases with age
- D. Asthma is usually disabling rather than lethal

✓ Answer is C

Explanation of Choices:

▼ Choice C is correct (the exception):

👉 Atopic asthma (the most common form) typically begins in childhood and is triggered by environmental allergens. Many children diagnosed with asthma experience spontaneous remission or significant improvement as they grow into adolescence or young adulthood; therefore, the severity of asthma does not typically increase with age.

▼ Choice A is incorrect:

👉 A defining characteristic of bronchial asthma is that the airway obstruction (bronchospasm) is intermittent and reversible (either spontaneously or with the help of bronchodilators).

▼ Choice B is incorrect:

👉 Asthma is characteristically an episodic disease marked by recurrent attacks of wheezing, chest tightness, and dyspnea interspersed with symptom-free intervals.

▼ Choice D is incorrect:

👉 In the vast majority of cases, asthma is a chronic, disabling nuisance that limits daily activities but is highly manageable; it only rarely becomes lethal in its most severe, unrelieved form (status asthmaticus).

Choice E is incorrect because Choice C is a valid exception.

📄 Core Pathology Note:

📖 Bronchial Asthma

Hallmarks:

- 🧬 Reversible bronchospasm, 🧬 chronic eosinophilic inflammation,
- 🧬 smooth muscle hypertrophy, and
- 🧬 mucous gland hyperplasia.

👃 Sputum Findings:

☞Curschmann spirals (mucus plugs) and

☞Charcot-Leyden crystals (composed of eosinophilic galectin-10).

18. Which type of atelectasis is irreversible?

- A. Contraction atelectasis
- B. Compression atelectasis
- C. Resorption atelectasis
- D. All

Answer: A. Contraction atelectasis

Explanation:

📶 Contraction atelectasis occurs when localized or generalized pulmonary fibrosis prevents full expansion of the lung. Because the underlying fibrosis and scarring permanently distort the lung architecture, this type is generally irreversible.

📶 Resorption atelectasis occurs when an airway is completely obstructed, preventing air from reaching the affected alveoli. Examples include mucus plugs, tumors, and foreign bodies. If the obstruction is removed, the lung can potentially re-expand.

✳️ Compression atelectasis occurs when the lung is compressed from outside by fluid, air, or a mass within the pleural cavity.

High-yield points:

▼ Contraction atelectasis → fibrosis/scarring → irreversible

▼ Resorption atelectasis → airway obstruction

▼ Compression atelectasis → external pressure on lung

Pneumothorax → compression atelectasis

19. Which one is not true about laryngeal cancer?

A. The most common histologic type is squamous cell carcinoma

B. The tumor usually develops on the vocal cords

C. More common in female

D. It is related to smoking and alcohol use

Answer: C. More common in female

Explanation:

▼ Laryngeal cancer is predominantly a disease of older men, although the incidence in women has increased with changing smoking patterns.

✓ Approximately the great majority of laryngeal carcinomas are squamous cell carcinomas.

✓ Smoking and alcohol are the major established risk factors, and their effects are synergistic.

👉 Tumors involving the true vocal cords often present early with persistent hoarseness, allowing earlier detection compared with tumors arising in less symptomatic locations.

★ High-yield points:

💡 Laryngeal cancer → usually squamous cell carcinoma

Major risk factors → smoking + alcohol

👤 More common in males

Persistent hoarseness → important warning sign

20. Which one is a complication of streptococcal tonsillopharyngitis?

- A. Acute rheumatic fever
- B. Acute glomerulonephritis
- C. Abscess
- D. A and C
- E. All

Answer: E. All

Explanation:

▼ Streptococcal tonsillopharyngitis, particularly infection caused by Group A β -hemolytic Streptococcus (*Streptococcus pyogenes*), can produce both local and systemic complications.

▼ Acute rheumatic fever is an immune-mediated complication that can occur after untreated or inadequately treated pharyngeal infection.

▼ It can affect the heart, joints, skin, and central nervous system.

▼ Poststreptococcal glomerulonephritis results from immune complex deposition and can cause hematuria, edema, hypertension, and renal dysfunction.

▼ Local suppurative complications can also occur, including peritonsillar abscess and other deep neck infections.

★ High-yield points:

🏴‍☠️ Group A Streptococcus → acute rheumatic fever

Group A Streptococcus → poststreptococcal glomerulonephritis

🏴‍☠️ Local complication → peritonsillar abscess

Rheumatic fever is immune-mediated; glomerulonephritis is immune-complex mediated

21. Which one factor affects the origin of nasopharyngeal carcinoma?

A. Age

B. Heredity

C. Infection with Epstein-Barr virus

D. All

Answer: D. All

Explanation:

▼ The development of nasopharyngeal carcinoma (NPC) is influenced by multiple factors rather than a single cause.

▼ Epstein-Barr virus (EBV) has a strong association with nasopharyngeal carcinoma, particularly with non-keratinizing and undifferentiated forms.

▼ Genetic and hereditary factors also contribute to susceptibility. Geographic and ethnic patterns further demonstrate the importance of genetic and environmental influences.

▼ Age also influences disease occurrence, with different age distributions depending on the population and geographic region.

★ High-yield points:

✓NPC → strongly associated with EBV, Genetic susceptibility contributes, Geographic distribution is characteristic

▼ NPC is particularly prevalent in certain Asian and African populations

22. Which one is not true about acute otitis media?

A. It is common in adults

B. Streptococcus pneumoniae is the most common bacterial agent

C. It is a disease of the middle ear

D. None

Answer: A. It is common in adults

Explanation:

- ▼ Acute otitis media is predominantly a disease of infants and young children, rather than adults.
- ▼ Children are particularly susceptible because their Eustachian tubes are shorter, more horizontal, and function less effectively, facilitating the spread of microorganisms from the nasopharynx into the middle ear.
- ▼ The infection involves the middle ear cavity and commonly follows an upper respiratory tract infection.
- ▼ Among bacterial causes, *Streptococcus pneumoniae* is an important and classic pathogen.

★ High-yield points:

▼ Acute otitis media → mainly infants and children

📌 Site → middle ear

🦠 Important bacterial cause → *Streptococcus pneumoniae*

👉 Eustachian tube anatomy predisposes children to infection

23. Which one is the commonest cause of tonsillopharyngitis?

- A. Viral
- B. Bacterial
- C. Fungi
- D. Parasitic

Answer: A. Viral

Explanation:

🌿 Viral infections are the most common cause of tonsillopharyngitis.

🌿 Common viral causes include rhinoviruses, adenoviruses, influenza viruses, parainfluenza viruses, and other respiratory viruses.

🌿 Viral infections commonly produce sore throat, fever, cough, rhinorrhea, and other upper respiratory symptoms.

🌿 Bacterial pharyngitis, particularly Group A Streptococcus, is clinically important because of its potential complications, but it accounts for a smaller proportion of cases than viral infection.

★ High-yield points:

📌 Most common cause of tonsillopharyngitis = viral

📌 Group A Streptococcus = important bacterial cause

📌 Viral pharyngitis commonly accompanies upper respiratory infection

24. Which one of the following is not a cause of resorption atelectasis?

- A. Foreign body aspiration
- B. Bronchial asthma
- C. Tumors
- D. Pneumothorax

Answer: D. Pneumothorax

Explanation:

▼ Resorption atelectasis occurs when an airway is completely obstructed.

▼ Air distal to the obstruction is gradually absorbed into the blood, causing the affected alveoli to collapse.

✓Common causes include foreign bodies, mucus plugs, and obstructing tumors.

✓Severe bronchial obstruction associated with asthma can also contribute to airway obstruction and atelectasis.

▼ Pneumothorax, however, does not cause resorption atelectasis. Air enters the pleural cavity and increases pressure around the lung, producing compression atelectasis.

▼ A useful anatomical distinction is that in resorption atelectasis, the mediastinum tends to shift toward the affected side, whereas significant compression atelectasis can shift it away from the affected lung.

★ High-yield points:

▼ Resorption atelectasis → airway obstruction

Causes → mucus plug, foreign body, tumor

▼ Pneumothorax → compression atelectasis

▼ Resorption → mediastinum shifts toward affected side

25. False about obstructive airway diseases?

A. Total lung capacity and forced vital capacity are normal

B. FEV1:FVC ratio is reduced

C. They are characterized by reduced expansion of lung parenchyma

D. They are caused by partial or complete obstruction at any level

Answer: C. They are characterized by reduced expansion of lung parenchyma

Explanation:

▼ Statement C describes the basic mechanism of restrictive lung disease, not obstructive lung disease.

👉 Obstructive diseases are characterized primarily by increased resistance to airflow, particularly during expiration.

▼ Examples include asthma, chronic bronchitis, and emphysema.

📌 The hallmark pulmonary function abnormality is a reduced FEV1/FVC ratio, because the patient has difficulty rapidly exhaling air.

▼ In restrictive disease, the major problem is reduced lung expansion, producing reduced lung volumes, particularly TLC.

★ High-yield points:

▼ Obstructive disease → difficulty getting air out

FEV1/FVC → decreased

▼ Restrictive disease → difficulty expanding lungs

Obstructive examples → asthma, chronic bronchitis, emphysema

26. Which of the following clinical features are not seen in patients with emphysema?

A. Barrel chest

B. Prominent hyperventilation

C. Cyanosis

D. Hunched-over positioning

Answer: C. Cyanosis

Explanation:

- ▼ The classic clinical description of a patient with predominant emphysema is the "pink puffer."
- ▼ These patients compensate for impaired gas exchange by increasing ventilation.
- ▼ They often have dyspnea, prolonged expiration, hyperinflation, and a barrel-shaped chest.
- ▼ They may lean forward and use accessory muscles to improve respiratory mechanics.
- ▼ Significant cyanosis is more characteristic of patients with predominant chronic bronchitis, traditionally described as "blue bloaters."

However, this distinction is a classic teaching description rather than an absolute clinical rule, because many patients have overlapping features of emphysema and chronic bronchitis.

★ High-yield points:

- ▼ Emphysema → "pink puffer"
- ▼ Chronic bronchitis → "blue bloater"
- ▼ Emphysema → hyperinflation + barrel chest + dyspnea
- ▼ Cyanosis is more characteristic of advanced chronic bronchitis

27. Diffuse alveolar damage is the pathologic manifestation of which clinical entity?

- A. Acute respiratory distress syndrome (ARDS)
- B. Pneumonia
- C. Pulmonary hemorrhage
- D. Pulmonary edema
- E. Pulmonary emboli

Answer: A. Acute respiratory distress syndrome (ARDS)

Explanation:

▼ Diffuse alveolar damage (DAD) is the characteristic histopathologic pattern associated with ARDS.

▼ ARDS results from severe injury to the alveolar-capillary barrier.

▼ Increased permeability allows protein-rich fluid to enter the alveolar spaces, producing pulmonary edema that is not primarily caused by left-sided heart failure.

▼ Histologically, the acute phase is characterized by capillary congestion, interstitial and alveolar edema, and formation of hyaline membranes.

▼ These hyaline membranes consist largely of fibrin-rich material and cellular debris lining the alveolar walls.

★ High-yield points:

🎯 ARDS → diffuse alveolar damage

🎯 DAD → alveolar edema + hyaline membranes

🎯 ARDS is a form of non-cardiogenic pulmonary edema

🎯 Major mechanism → diffuse alveolar-capillary injury

28. The Azzopardi effect is seen in which lung neoplasm?

- A. Squamous cell carcinoma
- B. Adenocarcinoma
- C. Small cell carcinoma
- D. Bronchioloalveolar carcinoma

Answer: C. Small cell carcinoma

Explanation:

▼ The Azzopardi effect is a characteristic microscopic finding associated with small cell carcinoma, particularly small cell carcinoma of the lung.

▼ It results from deposition of DNA from necrotic tumor cells around the walls of small blood vessels.

🔬 Microscopically, the basophilic nuclear material can appear to outline or coat vessel walls.

▪ Small cell carcinoma is composed of small malignant cells with scant cytoplasm, finely granular chromatin, nuclear molding, high mitotic activity, and extensive necrosis.

★ High-yield points:

▼ Azzopardi effect → small cell carcinoma

▼ Mechanism → deposition of nuclear DNA around vessel walls

✓ SCLC → nuclear molding + scant cytoplasm + high mitotic rate + necrosis

✓ SCLC → early metastasis and strong association with smoking

29. Which of the following is the most common site for the development of malignant mesothelioma?

A. Pleura

B. Peritoneum

C. Pericardium

D. Tunica vaginalis of testis

E. Mediastinum

Answer: A. Pleura

Explanation:

▼ The pleura is by far the most common site of malignant mesothelioma.

▼ Malignant mesothelioma is a tumor arising from mesothelial cells lining serous cavities. The strongest established etiologic association is occupational exposure to asbestos, particularly prolonged exposure involving construction, shipbuilding, insulation, and related industries.

▼ Pleural mesothelioma may present with chest pain, dyspnea, pleural effusion, and diffuse pleural thickening that can encase the lung.

★ High-yield points:

👉 Most common site of malignant mesothelioma → pleura

▼ Major risk factor → asbestos exposure

▼ Mesothelioma ≠ smoking-related cancer

▼ Typical presentation → pleural effusion + diffuse pleural thickening

30. Wegener granulomatosis can affect which of the following sites?

- A. Lung
- B. Kidney
- C. Peripheral nerves and central nervous system
- D. Skin and breast
- E. All

Answer: E. All

Explanation:

▼ Granulomatosis with polyangiitis (GPA), formerly called Wegener granulomatosis, is a systemic necrotizing granulomatous vasculitis that can involve multiple organs.

▼ The classic major organs involved are the upper respiratory tract, lungs, and kidneys.

▼ Pulmonary disease may produce nodules, cavitory lesions, pulmonary hemorrhage, and other abnormalities.

▼ Renal involvement typically manifests as rapidly progressive glomerulonephritis.

▼ Because GPA is a systemic vasculitis, other organs can also be affected, including the skin, peripheral nerves, eyes, and central nervous system.

▼ Therefore, among the options provided, all are possible sites of involvement.

GPA is classically associated with c-ANCA directed against proteinase-3 (PR3), although ANCA testing is not positive in every patient.

★ High-yield points:

▼ GPA → necrotizing granulomatous vasculitis

🧐 Classic triad → upper respiratory tract + lung + kidney

▼ Pulmonary lesions → nodules and cavitation

▼ Renal lesion → rapidly progressive glomerulonephritis

▼ Classic antibody → c-ANCA / PR3-ANCA

#2015 Pathology Exam

Question 1:

▼ Which is true about Resorption atelectasis?

A. leads to the resorption of oxygen trapped in the dependent alveoli

B. leads to mediastinal shift away from the affected lung

C. is often irreversible

D. lung fibrosis is a common cause

E. A and B are correct

✅ Answer is A

Explanation of Choices:

✅ Choice A is correct:

▼ Resorption (obstruction) atelectasis occurs when an airway is completely obstructed.

▼ Once obstruction is complete, the air (which contains oxygen) trapped in the distal alveoli gradually becomes absorbed (resorbed) into the surrounding capillary blood, leading to the collapse of the airspaces.

⊘ Choice B is incorrect: In resorption atelectasis, because lung volume is significantly diminished, the mediastinum shifts toward the collapsed lung.

↯ A mediastinal shift away from the affected lung is characteristic of compression atelectasis, which is caused by space-occupying fluid, blood, or air in the pleural cavity.

⊘ Choice C is incorrect:

↻ Atelectasis is generally a reversible disorder once the initiating obstruction or compressive force is resolved.

♣ The only form of atelectasis that is typically irreversible is contraction atelectasis.

⊘ Choice D is incorrect:

🫁 Lung or pleural fibrosis is the classic cause of contraction atelectasis (where fibrotic changes physically hamper lung expansion), not resorption atelectasis.

⊘ Resorption atelectasis is instead caused by mucus plugs, foreign body aspiration, tumors, or bronchial asthma.

Choice E is incorrect because Choice B is false.

Question 2:

▼ Which of the following statement is true?

A. Infection has no role in chronic bronchitis

B. Asthma is usually fatal

C. Smoking is a common risk factor for mesothelioma

D. Bronchiectasis more commonly affects the upper lobe of the lung

E. B and D are correct

F. None

✅ Answer is F (None)

Explanation of Choices:

❌ Choice A is incorrect:

👉 While cigarette smoke and air pollutants are the primary initiators of chronic bronchitis, infection plays a highly significant role in maintaining the chronic inflammation and is critical in triggering acute infectious exacerbations.

❌ Choice B is incorrect:

✦ Asthma is an episodic, manageable airway disease that is rarely fatal.

💀 It only becomes lethal in its most severe, unrelenting form—status asthmaticus—where extreme airflow obstruction causes severe cyanosis and potentially death.

❌ Choice C is incorrect:

↯ Cigarette smoking is not a risk factor for malignant mesothelioma.

↯ The overwhelming primary risk factor for mesothelioma of the pleura is asbestos exposure.

❌ Choice D is incorrect:

❧ Bronchiectasis characteristically and predominantly affects the lower lobes of the lungs
▼ bilaterally, being most severe in the distal bronchi and bronchioles.

✗ Choice E is incorrect because both Choices B and D are false.

✅ Choice F is correct because all statements (A, B, C, and D) are false.

✗ Choice E is incorrect because sarcoidosis typically exhibits a slight female predilection or is described as more common in young African American females and non-smokers.

✔ Choice F is correct because both Choices B and C are true.

Question 3:

▼ True about Sarcoidosis?

A. it mainly affects those older than 70 yrs

B. central necrosis is unusual

C. lymph nodes are involved in almost all case

D. asteroid bodies are pathognomonic

E. is more common in men

F. B and C are correct

✔ Answer is F (B and C are correct)

Explanation of Choices:

✗ Choice A is incorrect:

▼ Sarcoidosis characteristically affects a younger demographic, with most patients presenting under 40 years of age.

✓ Choice B is correct:

▼ Sarcoidosis is defined as a systemic inflammatory disease characterized by non-necrotizing (non-caseating) epithelioid granulomas.

Therefore, central caseating necrosis is absent and highly unusual.

✓ Choice C is correct:

✱ Lymph nodes are involved in almost all cases (75% to 90% or more), particularly the hilar and mediastinal lymph nodes, which present as bilateral hilar adenopathy on chest radiographs.

✗ Choice D is incorrect:

▼ While asteroid bodies (stellate inclusions within giant cells) and Schaumann bodies are often seen inside the granulomas, their presence is not pathognomonic, as they can also occur in granulomatous conditions of other origins. Like sporothrix

Question 4:

▼ From first to last, the morphological changes in lobar pneumonia occur in which correct chronological order?

A. grey hepatisation, red hepatisation, congestion, resolution

B. red hepatisation, grey hepatisation, congestion, resolution

C. congestion, grey hepatisation, red hepatisation, resolution

D. congestion, red hepatisation, grey hepatisation, resolution

E. congestion, red hepatisation, grey hepatisation, interstitial fibrosis.

✓ Answer is D

Explanation of Choices:

✓ Choice D is correct:

▼ The classic, untreated evolution of lobar pneumonia occurs in four well-defined chronological stages:

1. Congestion (Day 1–2):

▼ The lung is heavy, boggy, and red, characterized by vascular engorgement, intra-alveolar fluid with numerous bacteria, and rare neutrophils.

2. Red Hepatization (Day 2–4):

▼ The lung becomes firm, airless, and liver-like (hepatization), characterized by a massive confluent exudate of red blood cells, neutrophils, and fibrin packing the alveolar spaces.

3. Gray Hepatization (Day 4–8):

▼ The red blood cells progressively disintegrate (lyse), while a dry, firm, grayish-brown color develops due to the persistence of a fibrinosuppurative exudate.

4. Resolution (Day 8+):

▼ The intra-alveolar exudate is enzymatically digested to produce granular, semi-fluid debris that is resorbed, ingested by macrophages, or expectorated, restoring normal architecture.

Choices A, B, and C are incorrect because they list the stages out of their proper chronological order.

✗ Choice E is incorrect

✦ because lobar pneumonia typically resolves completely. It does not commonly progress to interstitial fibrosis unless rare organization of the exudate occurs into a fibrous scar.

Question 5:

▼ The most frequent cause of gram-negative bacterial pneumonia is?

A. *Streptococcus pneumoniae*

B. *Haemophilus influenzae*

C. *Moraxella catarrhalis*

D. *Staphylococcus aureus*

E. *Klebsiella pneumoniae*

✅ Answer is E

Explanation of Choices:

✅ Choice E is correct:

🔑 *Klebsiella pneumoniae* is the most frequent gram-negative bacterial pneumonia. It classically targets debilitated, malnourished, or chronic alcoholic populations and is characterized by a thick, gelatinous, viscid capsular sputum.

❌ Choice A is incorrect because *Streptococcus pneumoniae* is a Gram-positive lancet-shaped diplococcus.

❌ Choice B is incorrect: While *Haemophilus influenzae* is a Gram-negative bacterium that is a common cause of acute community-acquired typical pneumonias (especially in patients with COPD), it is not the most frequent or prototypical Gram-negative lobar pneumonia when compared to *Klebsiella*.

❌ Choice C is incorrect:

👉 *Moraxella catarrhalis* is a Gram-negative diplococcus that is a common cause of bacterial pneumonia in the elderly and patients with COPD, but it is less frequent than *Klebsiella*.

Choice D is incorrect because *Staphylococcus aureus* is a Gram-positive coccus.

Question 6:

▼ True about Squamous cell lung carcinoma?

A. has the poorest prognosis

B. is closely correlated with a smoking history

C. is commonest at the periphery of the lung

D. is more common in females

E. B and C are correct

✓ Answer is B

Explanation of Choices:

✓ Choice B is correct:

▼ Squamous cell carcinoma (SCC) of the lung is extremely closely and directly correlated with a history of cigarette smoking.

▼ Exposure to cigarette smoke drives a sequence of metaplastic and dysplastic changes in the bronchial epithelium that eventually lead to malignant SCC.

✗ Choice A is incorrect: The lung cancer with the absolute poorest prognosis is small cell lung carcinoma (SCLC), which metastasizes exceptionally early and has a rapid doubling time.

Question 7:

▼ Which one is not complication of bronchiectasis?

- A. Cor pulmonale
- B. Metastatic brain abscess
- C. Focus of repeated lung parenchyma infection
- D. Amyloidosis
- E. None

✓ Answer is E (None)

Explanation of Choices:

▼ Choice A is a complication:

★ Long-standing, severe bronchiectasis can lead to chronic hypoxia and widespread destruction of the pulmonary vascular bed.

★ This triggers persistent pulmonary arterial hypertension, which increases the workload on the right side of the heart, eventually resulting in right ventricular hypertrophy and right-sided heart failure (cor pulmonale).

Because this is a complication, it cannot be the correct answer to a "not" question.

▼ Choice B is a complication:

★ The chronically dilated bronchi in bronchiectasis accumulate pools of purulent secretions that harbor persistent infections.

✓ This tissue damage can allow bacteria to enter the systemic bloodstream (bacteremic dissemination) and seed distant organs, classically resulting in a metastatic brain abscess.

Because this is a complication, it is incorrect.

▼ Choice C is a complication:

👉 The permanent destruction of the bronchial elastic and muscular supporting tissue hampers normal clearance mechanisms. Consequently, the dilated airways serve as a constant reservoir for retained secretions and a focus of repeated, recurrent infections of the surrounding lung parenchyma. Because this is a complication, it is incorrect.

▼ Choice D is a complication:

★ The continuous, severe, localized tissue destruction in bronchiectasis represents a state of chronic, long-term inflammation. This persistent inflammatory stimulus can cause the systemic deposition of amyloid fibrils (AA amyloidosis), leading to organ dysfunction (secondary amyloidosis). Because this is a complication, it is incorrect.

Choice E is correct: Since all the listed conditions (A, B, C, and D) are well-established, classic pathological complications of bronchiectasis, "None" of the choices is a non-complication.

Question 8:

▼ Which disorder is characterized by permanent dilation of bronchi and bronchioles?

- A. Chronic bronchitis
- B. Emphysema
- C. Bronchiectasis
- D. Atelectasis

Answer: C. Bronchiectasis

Explanation:

▼ Bronchiectasis is characterized by permanent and irreversible dilation of bronchi and bronchioles resulting from destruction of the muscular and elastic components of the bronchial wall.

🔥 The underlying process is usually chronic inflammation and infection, often associated with impaired clearance of respiratory secretions.

✳️ Persistent infection damages the bronchial wall, producing progressive dilation and accumulation of purulent secretions.

⚡ Important causes include cystic fibrosis, bronchial obstruction, recurrent pulmonary infections, and congenital disorders affecting mucociliary clearance.

⚡ Clinically, patients commonly present with chronic cough and copious, purulent, sometimes foul-smelling sputum.

★ High-yield points:

✳️ Bronchiectasis = permanent bronchial dilation

⚡ Major mechanism = destruction of bronchial wall by chronic infection/inflammation

⚡ Classic presentation = chronic cough + copious purulent sputum

✳️ Cystic fibrosis is an important cause

👤 Bronchiectasis is defined as the permanent dilation of bronchi and bronchioles caused by the destruction of muscle and elastic supporting tissue.

▼ Bronchiectasis is usually secondary to chronic necrotizing infections or persistent obstruction.

Question :9

▼ A patient receives cardiopulmonary resuscitation and suffers aspiration of gastric contents. Over the next 10 days he develops a non-productive cough along with a fever to 37.9°C. A chest radiograph reveals a 4 cm diameter mass with an air-fluid level in the right lung. A culture done for the sputum shows mixed flora. Which of the following conditions is he most likely to have?

A. Squamous cell carcinoma

B. Lung abscess

C. Chronic bronchitis

D. Bronchiectasis

Answer: B. Lung abscess

Explanation:

🔄 The patient has several classic features of an aspiration-related lung abscess.

⚡ Aspiration of gastric or oropharyngeal contents can introduce bacteria into the lower respiratory tract. The resulting infection causes localized suppurative destruction of lung tissue, producing a cavity filled with pus.

🧠 The chest radiograph showing a cavity containing an air-fluid level is highly characteristic of a lung abscess. The mixed bacterial flora also supports aspiration because aspiration-related abscesses are frequently caused by mixed aerobic and anaerobic organisms.

🌐 The right lung is commonly affected because the right main bronchus is wider, shorter, and more vertically oriented than the left main bronchus.

★ High-yield points:

▼ Aspiration → common cause of lung abscess

▼ Lung abscess → localized suppurative necrosis

▼ Air-fluid level → classic radiologic finding

Aspiration abscess → often mixed bacterial/anaerobic flora

▼ Right lung is commonly affected

Question :10

▼ A 30-year-old male with cystic fibrosis presents with shortness of breath and production of a large amount of foul-smelling sputum. What is the most likely cause?

- A. Atelectasis
- B. Asthma
- C. Bronchiectasis
- D. Adenocarcinoma

Answer: C. Bronchiectasis

Explanation:

▼ Cystic fibrosis causes abnormal production of thick, dehydrated mucus because of defective CFTR-mediated chloride transport.

▼ The thick secretions impair mucociliary clearance and predispose the respiratory tract to recurrent bacterial infections.

▼ Repeated infection and inflammation progressively destroy the muscular and elastic components of the bronchial walls, resulting in bronchiectasis.

▼ Bronchiectasis produces dilated airways that retain large amounts of infected secretions.

▼ Therefore, patients commonly develop chronic productive cough with copious purulent or foul-smelling sputum, recurrent infections, dyspnea, and sometimes hemoptysis.

★ High-yield points:

🔄 Cystic fibrosis → thick mucus → impaired mucociliary clearance → recurrent infection → bronchiectasis

Bronchiectasis → copious purulent/foul-smelling sputum

▼ *Pseudomonas aeruginosa* is an important pathogen in CF.

Question 11:

▼ The most common outcome of treated bacterial pneumonia is?

A. Resolution

B. Scar formation

C. Abscess formation

D. Bronchopleural fistula

Answer: A. Resolution

Explanation:

▼ The most common outcome of appropriately treated bacterial pneumonia is complete resolution.

🧬 During resolution, inflammatory exudate within the alveolar spaces is enzymatically digested.

▼ Macrophages remove cellular debris, and the remaining material is either expectorated or drained through lymphatic pathways.

▼ Because the basic alveolar framework is usually preserved, the lung can return to normal architecture without significant scarring.

▼ Complications such as abscess formation, empyema, organization, bacteremia, and bronchopleural fistula can occur but are much less common.

★ High-yield points:

✦ Most common outcome of bacterial pneumonia = resolution

✦ Pneumonia usually heals without significant fibrosis

Complications include abscess, empyema, bacteremia, and organization

Question: 12

▼ A 60-year-old male with a history of smoking presents with hoarseness of voice. Laryngoscopy reveals a 3 cm laryngeal mass. On histopathologic examination, the mass was identified as malignant neoplasm. The most likely histologic diagnosis is?

A. Adenocarcinoma

B. Small cell carcinoma

C. Squamous cell carcinoma

D. Lymphoma

Answer: C. Squamous cell carcinoma

Explanation:

▼ The overwhelming majority of laryngeal malignant tumors are squamous cell carcinomas.

▼ The major risk factors are cigarette smoking and alcohol consumption, with the risk increasing substantially when both are present.

📌 Laryngeal SCC commonly occurs in older men.

🐎 Hoarseness is particularly important because tumors involving the true vocal cords interfere with vocal cord vibration.

▼ Persistent unexplained hoarseness in an older smoker should therefore raise suspicion for laryngeal carcinoma.

★ High-yield points:

▼ Most common laryngeal malignancy = squamous cell carcinoma

🎯 Major risk factors = smoking 🚬 + alcohol 🍷

▼ Hoarseness = classic early symptom, especially with vocal cord involvement

Persistent hoarseness in a smoker requires evaluation

Question: 13

▼ The most common causative organism for community-acquired typical pneumonia is?

- A. Streptococci
- B. Pneumococci
- C. Haemophilus
- D. Staphylococci

Answer: B. Pneumococci

Explanation:

▼ *Streptococcus pneumoniae* (pneumococcus) is the classic and most common cause of community-acquired typical bacterial pneumonia.

✱ Pneumococcal pneumonia classically produces an acute illness with fever, productive cough, pleuritic chest pain, and radiographic consolidation.

✓ Morphologically, it can produce the classical stages of lobar pneumonia: congestion, red hepatisation, grey hepatisation, and resolution.

★ High-yield points:

▼ Most common classic cause of community-acquired typical pneumonia = *Streptococcus pneumoniae*

🏠 Typical bacterial pneumonia → alveolar consolidation

Pneumococcus → classically associated with lobar pneumonia

Question:14

▼ Emphysema associated with alpha-1 antitrypsin deficiency is?

- A. Panacinar
- B. Distal acinar
- C. Centriacinar
- D. Irregular

Answer: A. Panacinar

Explanation:

▼ Panacinar emphysema is strongly associated with alpha-1 antitrypsin deficiency.

✦ Alpha-1 antitrypsin is a protease inhibitor  that normally protects lung tissue from enzymes such as neutrophil elastase.

 Deficiency allows excessive proteolytic destruction of alveolar walls.

 Panacinar emphysema involves the entire acinus relatively uniformly, from the respiratory bronchiole through the alveolar ducts and alveoli.

 It has a characteristic predominance in the lower lobes, especially in severe alpha-1 antitrypsin deficiency.

▲ In contrast, centriacinar emphysema is strongly associated with cigarette smoking and predominantly affects the upper lobes.

★ High-yield points:

⊘ Alpha-1 antitrypsin deficiency → panacinar emphysema

☞ Panacinar → lower lobes

🚬 Smoking → centriacinar emphysema → upper lobes

⊘ Alpha-1 antitrypsin normally inhibits neutrophil elastase.

Question 15

▼ Asbestos exposure results in all of the following changes except?

- A. Interstitial pneumonia
- B. Malignant mesothelioma
- C. Pleural effusion
- D. Pleural fibrosis

Answer: A. Interstitial pneumonia

Explanation:

▼ Asbestos exposure causes several characteristic pulmonary and pleural abnormalities, including asbestosis, pleural plaques, pleural fibrosis, pleural effusions, and malignant mesothelioma.

✓ The classic parenchymal lesion is interstitial pulmonary fibrosis, known as asbestosis.

✿ Therefore, "interstitial pneumonia" is not the appropriate characteristic asbestos-related lesion in this context.

▼ Asbestos fibers become coated with iron-containing material and proteins, producing characteristic asbestos bodies, which appear as golden-brown ●, beaded or dumbbell-shaped structures microscopically.

★ High-yield points:

○ Asbestos → interstitial pulmonary fibrosis (asbestosis)

○ Asbestos → malignant pleural mesothelioma

○ Asbestos → pleural plaques and fibrosis

● Asbestos bodies = golden-brown coated fibers

▲ Smoking + asbestos → markedly increased risk of bronchogenic carcinoma

Question 16

▼ The most common etiologic factor implicated in chronic bronchiolitis is?

- A. Air pollutant
- B. Mycoplasma infection
- C. Cigarette smoking
- D. Cotton mills

Answer: C. Cigarette smoking

Explanation:

©Cigarette smoking is the most important etiologic factor associated with chronic bronchitis and chronic small-airway disease, including chronic bronchiolitis.

+ Repeated exposure to tobacco smoke causes chronic inflammation of the bronchi and bronchioles, increased mucus production, mucus plugging, and structural changes in the airway walls.

↗ These changes increase airflow resistance and contribute to obstructive lung disease.

★ High-yield points:

▼ Cigarette smoking = major cause of chronic bronchitis and chronic bronchiolitis

✦ Chronic bronchiolitis → inflammation + mucus plugging + airway narrowing

Smoking is the major preventable risk factor for COPD

Question 17

▼ Among the following genes, which one is mostly affected in small cell lung carcinoma?

- A. RB gene
- B. CDK gene
- C. MYC gene
- D. RAS gene

Answer: A. RB gene

Explanation:

▼ Small cell lung carcinoma (SCLC) is strongly associated with inactivation of the RB1 and TP53 tumor suppressor genes.

▼ Loss of RB function removes an important checkpoint controlling progression through the cell cycle.

🧬 Loss of p53 further compromises DNA-damage responses and apoptosis.

✳️ These genetic abnormalities contribute to the highly aggressive behavior of SCLC.

🎯 SCLC is strongly associated with cigarette smoking, has a high mitotic rate, extensive necrosis, and tends to metastasize early.

★ High-yield points:

✦ SCLC → RB1 and TP53 abnormalities

✦ SCLC → strongly associated with smoking

✦ SCLC → aggressive tumor with early metastasis

✦ Azzopardi effect → classically associated with SCLC

Question 18

▼ Which of the following organisms is a common cause of pneumonia in patients with cystic fibrosis?

- A. Haemophilus influenzae
- B. Pseudomonas aeruginosa
- C. Legionella pneumophila
- D. Moraxella catarrhalis

Answer: B. Pseudomonas aeruginosa

Explanation:

▼ Patients with cystic fibrosis have thick, viscous respiratory secretions because of abnormal CFTR function.

▼ These secretions impair mucociliary clearance and create an environment favorable for persistent bacterial colonization.

🌿 *Pseudomonas aeruginosa* is particularly important in patients with established cystic fibrosis.

🌊 Chronic colonization can cause recurrent pulmonary infections and progressive structural lung damage, especially bronchiectasis.

★ High-yield points:

✂ Cystic fibrosis → *Pseudomonas aeruginosa* is a major pulmonary pathogen

✂ CF → impaired mucociliary clearance

🎯 Chronic *Pseudomonas* infection → recurrent pneumonia + bronchiectasis

Question 19

▼ Restrictive lung diseases are characterized by all of the following except?

- A. Shortness of breath
- B. Cyanosis
- C. Tachypnea
- D. Wheezing

Answer: D. Wheezing

Explanation:

✚ Restrictive lung diseases are characterized primarily by reduced lung volumes, particularly reduced total lung capacity.

✚ The lungs are difficult to expand because of either intrinsic parenchymal disease or extrapulmonary causes.

✚ Patients commonly present with dyspnea and tachypnea, and severe disease can result in hypoxemia and cyanosis.

✚ Wheezing, however, is more characteristic of obstructive airway disease, where narrowing of the airways produces turbulent airflow.

Examples include asthma and COPD.

🎯 In pulmonary function testing, restrictive disease typically shows reduced FVC and TLC with a normal or increased FEV1/FVC ratio.

★ High-yield points:

✓ Restrictive disease → ↓ TLC

FEV1/FVC → normal or increased

✳ Obstructive disease → ↓ FEV1/FVC

👉 Wheezing → more characteristic of obstructive disease

Question 20

▼ Lung cancer that produces hypercalcemia is mostly?

- A. Squamous cell carcinoma
- B. Adenocarcinoma
- C. Small cell carcinoma
- D. Carcinoid tumor

Answer: A. Squamous cell carcinoma

Explanation:

👉 Squamous cell carcinoma of the lung is classically associated with hypercalcemia as a paraneoplastic syndrome.

▼ The mechanism is usually production of parathyroid hormone-related peptide (PTHrP) by tumor cells.

🧠 PTHrP increases calcium levels by promoting bone resorption and increasing renal calcium retention.

▼ This is an important distinction among lung cancers. Squamous cell carcinoma is associated with PTHrP-mediated hypercalcemia, whereas small cell carcinoma is more strongly associated with SIADH and ectopic ACTH production.

★ High-yield points:

▼ Squamous cell carcinoma → PTHrP → hypercalcemia

▼ SCLC → SIADH and ectopic ACTH

Lung cancer can produce important paraneoplastic syndromes

Question 21

▼ The most common cause of death in Goodpasture syndrome is?

A. Respiratory failure

B. Uremia

C. Hemoptysis

D. None

Answer: B. Uremia

Explanation:

▼ Goodpasture syndrome is an autoimmune disease caused by antibodies directed against the glomerular and alveolar basement membranes.

▼ These antibodies recognize the alpha-3 chain of type IV collagen.

🔑 The disease produces a combination of rapidly progressive glomerulonephritis and pulmonary hemorrhage.

👉 Although pulmonary manifestations such as hemoptysis can be dramatic, progressive renal injury can result in severe renal failure and uremia, which is a major cause of death if untreated.

★ High-yield points:

✓ Goodpasture syndrome → anti-GBM antibodies

Target → type IV collagen in glomerular and alveolar basement membranes

👤 Clinical combination → pulmonary hemorrhage + rapidly progressive glomerulonephritis

📌 Renal failure → major cause of mortality

🔔 Correction

Question 22

▼ Which of the following is not a typical feature of asthma?

- A. Reversibility
- B. Asthma is recurrent
- C. The severity of asthma increases with age
- D. Asthma is usually disabling rather than lethal
- E. None

✓ Different sources give different answer but we chose the best one.

✅ Answer is C

Explanation of Choices:

▼ Choice C is correct (the exception):

👉 Atopic asthma (the most common form) typically begins in childhood and is triggered by environmental allergens. Many children diagnosed with asthma experience spontaneous remission or significant improvement as they grow into adolescence or young adulthood; therefore, the severity of asthma does not typically increase with age.

▼ Choice A is incorrect:

👉 A defining characteristic of bronchial asthma is that the airway obstruction (bronchospasm) is intermittent and reversible (either spontaneously or with the help of bronchodilators).

▼ Choice B is incorrect:

👉 Asthma is characteristically an episodic disease marked by recurrent attacks of wheezing, chest tightness, and dyspnea interspersed with symptom-free intervals.

▼ Choice D is incorrect:

👉 In the vast majority of cases, asthma is a chronic, disabling nuisance that limits daily activities but is highly manageable; it only rarely becomes lethal in its most severe, unrelieved form (status asthmaticus).

Choice E is incorrect because Choice C is a valid exception.

 Core Pathology Note:

 Bronchial Asthma

↔ Hallmarks:

 Reversible bronchospasm,  chronic eosinophilic inflammation,
 smooth muscle hypertrophy, and
 mucous gland hyperplasia.

🦠 Sputum Findings:

🌀 Curschmann spirals (mucus plugs) and

🌀 Charcot-Leyden crystals (composed of eosinophilic galectin-10).

Question 23:

▼ Not common cause of death in patients with emphysema?

A. coronary artery disease

B. respiratory failure

C. right-sided heart failure

D. massive collapse of the lung

E. None

✅ Answer is E (None)

Explanation of Choices:

▼ Choice E is correct:

👉 According to your respiratory pathology lecture slides on the clinical course of emphysema,

💀 Death in most patients with emphysema is due to:

🔴 Coronary artery disease (ischemic complications related to chronic hypoxemia and systemic stress).

🫁 Respiratory failure accompanied by respiratory acidosis, severe hypoxia, and coma.

💜 Right-sided heart failure (cor pulmonale) secondary to severe pulmonary hypertension.

🫁 Massive collapse of the lungs secondary to a spontaneous pneumothorax (often due to ruptured apical bullae/blebs).

😊 Since all of the choices (A, B, C, and D) are explicitly listed as major and common causes of death in emphysema patients, "None" of the listed options is a non-cause.

Question 24

▼ Which type of atelectasis is irreversible?

- A. Contraction atelectasis
- B. Compression atelectasis
- C. Resorption atelectasis
- D. All

Answer: A. Contraction atelectasis

Explanation:

📶 Contraction atelectasis occurs when localized or generalized pulmonary fibrosis prevents full expansion of the lung. Because the underlying fibrosis and scarring permanently distort the lung architecture, this type is generally irreversible.

📶 Resorption atelectasis occurs when an airway is completely obstructed, preventing air from reaching the affected alveoli. Examples include mucus plugs, tumors, and foreign bodies. If the obstruction is removed, the lung can potentially re-expand.

✳️ Compression atelectasis occurs when the lung is compressed from outside by fluid, air, or a mass within the pleural cavity.

High-yield points:

▼ Contraction atelectasis → fibrosis/scarring → irreversible

▼ Resorption atelectasis → airway obstruction

▼ Compression atelectasis → external pressure on lung

Pneumothorax → compression atelectasis

Question 25:

▼ Which of the following lung cancer has the poorest prognosis?

A. small cell carcinoma

B. adenocarcinoma

C. squamous cell carcinoma

D. large cell carcinoma

✅ Answer is A

Explanation of Choices:

▼ Choice A is correct:

▪ Small cell lung carcinoma (SCLC) has the poorest prognosis of all lung cancers.

🔴 SCLC is an extremely aggressive neuroendocrine malignancy that is highly linked to smoking.

📌 It is characterized by rapid doubling time, frequent vascular invasion, and near-universal metastasis at the time of diagnosis, making it typically unamenable to surgical resection.

▼ Choice B is incorrect:

👉 While adenocarcinoma is highly malignant and metastasizes widely, its progression is slower than SCLC, and early-stage peripheral adenocarcinomas are highly treatable with surgical resection, yielding a much better prognosis.

▼ Choice C is incorrect:

👉 Squamous cell carcinoma is typically central, grows relatively slowly, and tends to remain localized to the thoracic cavity for longer periods before metastasizing, giving it a better prognosis and higher surgical resectability rate compared to SCLC.

▼ Choice D is incorrect:

👉 Large cell carcinoma is an undifferentiated tumor with a poor prognosis and early metastasis, but it does not exhibit the extremely rapid, near-universal, early hematogenous spread seen in SCLC.

Question 26:

▼ Which of the following viruses is not commonly associated with the common cold?

A. coronavirus

B. adenovirus

C. rhinovirus

D. varicella-zoster virus

✅ Answer is D

Explanation of Choices:

▼ Choice D is correct:

✓ Varicella-zoster virus (VZV) is a herpesvirus responsible for causing chickenpox (varicella) and shingles (herpes zoster). It is not associated with upper respiratory tract catarrhal infections like the common cold.

▼ Choice A is incorrect:

👉 Coronaviruses are classic, common viral agents responsible for a significant percentage of common colds in humans, producing typical catarrhal upper airway symptoms.

▼ Choice B is incorrect:

👉 Adenoviruses are well-known causes of acute upper respiratory infections, presenting classically with pharyngotonsillitis, conjunctivitis, and rhinitis.

▼ Choice C is incorrect:

👉 ★ Rhinoviruses are the single most common cause of the common cold (infectious rhinitis), binding to ICAM-1 receptors on respiratory epithelial cells.

Question 27:

▼ Which of the following gram-negative aerobic bacteria causes one of the major respiratory tract infections, most commonly the nasopharynx, and is responsible for the infection of "otitis media" in children?

- A. *Bordetella pertussis*
- B. *Pseudomonas aeruginosa*
- C. *Moraxella catarrhalis*
- D. *Yersinia pestis*

✅ Answer is C

Explanation of Choices:

▼ Choice C is correct:

✦ *Moraxella catarrhalis* is a Gram-negative aerobic diplococcus ●● that commonly colonizes the nasopharynx. 📌

✓ It is a well-established major pathogen responsible for causing otitis media (middle ear infection) and sinusitis in infants and children, as well as acute exacerbations of COPD in adults.

▼ Choice A is incorrect:

✓ *Bordetella pertussis* is a Gram-negative coccobacillus that causes whooping cough (pertussis), characterized by severe paroxysmal coughing fits, but it is not a primary or common cause of acute suppurative otitis media.

▼ Choice B is incorrect:

✓ *Pseudomonas aeruginosa* is a Gram-negative rod that is associated with chronic otitis media and necrotizing otitis externa (especially in diabetics), but it is not the classic, primary cause of acute community-acquired otitis media in children.

▲ Choice D is incorrect:

✓ *Yersinia pestis* is a Gram-negative coccobacillus that causes plague (bubonic, septicemic, or pneumonic), which is a rare, highly severe zoonotic infection not related to routine childhood middle ear disease.

▲ Major Pathogens:

▪ Acute otitis media in children is most commonly caused by

▪ *Streptococcus pneumoniae*, ▪ non-typeable *Haemophilus influenzae*, and

▪ *Moraxella catarrhalis*.

Question 28:

▼ Which of the following pathogen is the most common cause of "bronchiolitis" in infants?

A. Respiratory Syncytial virus

B. Herpes simplex virus

C. Coxsackievirus

D. Rhinovirus

✅ Answer is A

Explanation of Choices:

▼ Choice A is correct

👉 (Respiratory Syncytial Virus - RSV): RSV is the single most common cause of acute bronchiolitis in infants and young children.

👉 It infects the small airways, leading to necrosis of bronchiolar epithelium, mucus plugging of the bronchiolar lumens, and subsequent small airway obstruction.

▼ Choice B is incorrect:

👉 Herpes simplex virus (HSV) typically causes vesicular lesions of the oral cavity (gingivostomatitis) or skin, rather than acute bronchiolitis in infants.

▼ Choice C is incorrect:

👉 Coxsackievirus is commonly associated with herpangina and hand, foot, and mouth disease, but is not a primary driver of infant bronchiolitis.

▼ Choice D is incorrect:

👉 Rhinovirus is the primary etiologic agent of the common cold (infectious rhinitis), which is an upper respiratory tract infection that does not typically involve the distal bronchioles of infants.

Question 29:

▼ All of the following statements are correct about the characteristics of chronic "sinusitis", Except?

A. Cold and cough that lasts for more than a month can cause viral or bacterial sinusitis

B. The infection can occur in all ages

C. Common cold, smoking, and seasonal allergies are the common risk factors

D. Pseudomonas aeruginosa is the most common bacteria that cause sinusitis

✅ Answer is D

Explanation of Choices:

▼ Choice D is correct (the exception):

👉 Pseudomonas aeruginosa is not the most common bacterial cause of sinusitis.

★ Acute and chronic sinusitis are typically caused by extension of infections from the nasal mucosa, driven by the normal inhabitants of the oral cavity (such as Streptococcus pneumoniae, Haemophilus influenzae, and Moraxella catarrhalis), and chronic forms frequently involve mixed anaerobic bacteria. (Pseudomonas aeruginosa is instead a common cause of chronic ear infections—chronic otitis media—but not chronic sinusitis).

▼ Choice A is incorrect (true statement):

★ A long-standing viral upper respiratory infection (cold and cough) that persists can lead to mucosal edema, blocking sinus drainage ostia and facilitating bacterial superinfection and chronic sinusitis.

▼ Choice B is incorrect (true statement):

👉 Sinusitis is a common inflammatory disease that can affect individuals across all age groups.

▼ Choice C is incorrect (true statement):

👉 The common cold (infectious rhinitis), cigarette smoking (which damages ciliary clearance), and seasonal allergies (allergic rhinitis) are well-known predisposing risk factors that induce mucosal edema and impair sinus drainage.

Question 30:

▼ A 4-year-old child developed symptoms of a common cold, after 2 days the cough started sounding similar to an animal bark. He was taken to the hospital on the third day with persistent fever and louder breathing sounds. The area below the chin appear to be swollen during the physical examination, which of the following could be the possible pathogen?

A. *Haemophilus influenzae*

B. Rhinovirus

C. Parainfluenza virus

D. Mycobacterium tuberculosis

✅ Answer is C

Explanation of Choices:

▼ Choice C is correct (Parainfluenza virus):

★ A harsh cough sounding like an "animal bark" 🐎 🦁 accompanied by persistent fever and inspiratory stridor in a young child is the classic clinical presentation of Croup (laryngotracheobronchitis).

★ The most common viral pathogen responsible for Croup is Parainfluenza virus.

▼ Choice A is incorrect:

★ Haemophilus influenzae is the primary cause of acute laryngo-epiglottitis in children.

✖ Epiglottitis is characterized by severe dysphagia, drooling, and a muffled "hot potato" voice, but it does not present with the classic barking cough of croup.

▼ Choice B is incorrect:

👉 Rhinovirus causes the common cold (infectious rhinitis) presenting with rhinitis, nasal congestion, and mild sore throat, but does not typically cause the subglottic narrowing that produces a barking cough.

▼ Choice D is incorrect:

★ Mycobacterium tuberculosis causes a chronic granulomatous infection (tuberculosis) presenting with productive cough, night sweats, and weight loss, and does not cause acute croup in pediatric patients.

Question 31:

▼ What is one key anatomic landmark that separates the upper respiratory tract from the lower respiratory tract?

A. pharynx

B. larynx

C. trachea

D. main bronchi

✓ Answer is B

Explanation of Choices:

▼ Choice B is correct (Larynx):

▪Anatomically, the larynx (voice box) is the transitional landmark that separates the upper respiratory tract (nose, nasal cavity, paranasal sinuses, and pharynx) from the lower respiratory tract (which begins at the trachea and extends to the lungs).

▼ Choice A is incorrect:

👉 The pharynx is located entirely within the upper respiratory tract, superior to the larynx.

▼ Choice C is incorrect:

👉 The trachea is the most superior structure of the lower respiratory tract, located below the transitional boundary of the larynx.

▼ Choice D is incorrect:

★ The main bronchi branch off from the bifurcation of the trachea deep within the lower respiratory tract.

Question 32:

▼ The deadliest and life threatening upper respiratory tract infection that needs emergency intervention is

A. tonsilopharyngitis

B. croup

C. epiglottitis

D. mastoiditis

✅ Answer is C

Explanation of Choices:

▼ Choice C is correct (Epiglottitis):

👉 Epiglottitis (specifically acute laryngo-epiglottitis) causes rapid, severe inflammatory swelling of the epiglottis and vocal cords. Because of the small size of pediatric airways, this swelling can cause sudden, complete laryngeal obstruction, constituting a life-threatening medical emergency that requires immediate airway intervention.

▼ Choice A is incorrect:

👉 Tonsillopharyngitis (sore throat) is a common, usually self-limiting upper respiratory infection that rarely compromises the airway.

▼ Choice B is incorrect:

👉 While Croup causes subglottic narrowing and inspiratory stridor, it is typically a self-limiting infection that is managed conservatively with steroids, and rarely carries the same high risk of sudden, total, fatal airway occlusion as acute epiglottitis.

▼ Choice D is incorrect:

👉 Mastoiditis is a suppurative infection of the mastoid air cells, typically complicating otitis media in the ear. While serious, it does not cause acute upper airway occlusion.

Question 33:

▼ Which of the following neoplasm is closely associated with EBV infection?

- A. Laryngeal papilloma
- B. Nasopharyngeal carcinoma
- C. Bronchoalveolar carcinoma
- D. Carcinoid tumor

✅ Answer is B (Nasopharyngeal carcinoma)

Explanation of Choices:

▼ Choice B is correct (Nasopharyngeal carcinoma):

★ Nasopharyngeal carcinoma (NPC) shows a strong, well-established oncogenic association with Epstein-Barr Virus (EBV) infection.

👉 EBV infects the host by first replicating in the nasopharyngeal mucosal epithelium and then infecting nearby tonsillar B lymphocytes, leading to clonal transformation.

✚ Unlike other EBV-associated malignancies (such as Burkitt lymphoma, where association varies), the EBV genome is found in virtually 100% of nasopharyngeal carcinomas (including non-endemic cases).

▼ Choice A is incorrect:

★ Laryngeal papillomas are

✓ benign,

✓raspberry-like exophytic vocal cord neoplasms

✓caused exclusively by Human Papillomavirus (HPV) types 6 and 11 acquired via the maternal birth canal during birth.

▼ Choice C is incorrect:

★ Bronchoalveolar carcinoma (now classified under adenocarcinoma in situ of the lung) is a primary pulmonary neoplasm closely linked to tobacco smoke exposure or driver oncogene mutations (such as EGFR mutations or ALK translocations in non-smokers), and is not associated with EBV.

▼ Choice D is incorrect:

👉 Bronchial carcinoids are primary neuroendocrine lung tumors thought to arise from the Kulchitsky (neuroendocrine) cells lining the bronchial mucosa. They are not associated with viral infections like EBV.

Question 34:

▼ The histological findings in asthma include, all EXCEPT:

A. A thinning of the basement membrane of bronchial epithelium

B. Oedema and an inflammatory infiltrate in bronchial walls with a predominance of neutrophils

C. An increase in size of sub mucosal glands

D. Hypertrophy of bronchial wall muscle

▼ Answer is A (A thinning of the basement membrane of bronchial epithelium) (Note: Choice B is also histologically atypical as asthma is classically characterized by an eosinophilic and mast cell infiltrate, but Choice A is a direct, absolute anatomical opposite of a major defining diagnostic criteria of asthma, making it the most definitive "all EXCEPT" selection).

Explanation of Choices:

▼ Choice A is correct (the exception):

▼ On microscopic examination, a classic diagnostic hallmark of bronchial asthma is marked thickening of the subepithelial basement membrane (due to subepithelial collagen deposition).

✓ Therefore, stating that asthma is characterized by "thinning" of the basement membrane is completely incorrect.

▼ Choice B is not a classic finding (but less of a direct anatomical opposite than A):

★ The prototypical inflammatory infiltrate in asthma is rich in eosinophils, mast cells, and CD4+ Th2 lymphocytes rather than neutrophils. However, neutrophils can sometimes be seen in severe acute asthma exacerbations or late-stage remodeling, whereas thinning of the basement membrane never occurs.

▼ Choice C is a classic finding (thus incorrect as the exception):

☞ Asthma features chronic mucus hypersecretion, which histologically corresponds to hyperplasia and an increase in the size of the submucosal mucous glands.

▼ Choice D is a classic finding (thus incorrect as the exception):

👉 Chronic bronchial hyperreactivity and persistent bronchospasm in asthma lead to prominent hypertrophy of the bronchial wall smooth muscle.

Question 35:

▼ Which one is not predisposing factor for bacterial sinusitis?

- A. nasal and sinus tumors
- B. granulomatous diseases (e.g., Wegener's granulomatosis or rhinoscleroma)
- C. altered mucus content (e.g., cystic fibrosis)
- D. In the hospital setting, nasotracheal intubation
- E. None

✅ Answer is E (None)

▼ Explanation of Choices:

▼ Choice E is correct:

👉 All of the listed choices (A, B, C, and D) are well-established predisposing risk factors that block sinus drainage ostia, leading to fluid retention and subsequent bacterial sinusitis:

▼ A is a predisposing factor: Nasal and sinus tumors (e.g., sinonasal papillomas or angiofibromas) physically obstruct sinus drainage tracts, causing retention of mucous secretions and secondary bacterial infection.

▼ B is a predisposing factor:

✚ Granulomatous diseases (such as Wegener's granulomatosis or rhinoscleroma) induce chronic mucosal damage, ulcerative lesions, and tissue remodeling, leading to chronic sinusitis.

▼ C is a predisposing factor:

✚ Cystic fibrosis alters chloride transport, producing abnormally thick, viscid mucus that plugs the sinuses and impairs ciliary clearance, predisposing to recurrent infections.

▼ D is a predisposing factor:

✓ In hospitalized or ICU patients, nasotracheal intubation blocks normal nasal airway passages and sinus ostia drainage, acting as a major trigger for severe nosocomial bacterial sinusitis.

Since all choices are valid predisposing factors, "None" of the above is a non-predisposing factor.

Question 36:

▼ Regarding emphysema:

A. the types include centriacinar, paraseptal, periseptal and irregular

B. the disease is caused by an excess of α_1 antitrypsin

C. the pattern is centriacinar in smokers

D. the protease- antiprotease hypothesis has been proven incorrect

E. the bullae or blebs are <1cm diameter

✅ Answer is C (the pattern is centriacinar in smokers)

Explanation of Choices:

▼ Choice c is correct:

🎯 Centriacinar (centrilobular) emphysema selectively destroys the central/proximal parts of the acinus (respiratory bronchioles) while sparing distal alveoli. This pattern is highly characteristic of cigarette smoking and predominantly affects the upper lobes of the lungs.

▼ Choice a is incorrect:

★ The four distinct morphological types of emphysema are: centriacinar, panacinar, distal acinar (paraseptal), and irregular. "Periseptal" is not a recognized pathological type.

▼ Choice b is incorrect:

★ Emphysema is associated with a genetic deficiency (not an excess) of the antiprotease alpha-1 antitrypsin (A1AT).

▼ Choice d is incorrect: The protease-antiprotease hypothesis remains the universally accepted pathogenetic model for emphysema. It states that tissue destruction arises from an imbalance between elastases (secreted by neutrophils and macrophages) and protective antiproteases like A1AT.

▼ Choice e is incorrect: In advanced emphysema, large subpleural air-filled spaces form. By definition, blebs or bullae are larger than 1 cm in diameter (bullae are >1 cm; those <1 cm are not classified as such).

Question 37:

▼ All of the following are neoplastic syndromes associated with lung cancer except

A. Cushing's

B. SIADH

C. Hypocalcaemia

D. Carcinoid

E. Hypertrophic osteoarthropathy

✓ Answer is C (Hypocalcaemia)

Explanation of Choices:

▼ Choice c is the correct exception (thus the correct answer): Lung cancer—specifically squamous cell carcinoma (SCC)—is notoriously associated with hypercalcemia (abnormally elevated blood calcium levels).

🎯 This is due to the ectopic secretion of parathyroid hormone-related protein (PTHrP). It does not cause hypocalcemia (low calcium), making this choice the absolute exception.

▼ Choice a is incorrect as an exception:

★ Cushing's syndrome is a classic paraneoplastic syndrome caused by the ectopic secretion of ACTH (adrenocorticotrophic hormone), which is highly characteristic of small cell lung carcinoma (SCLC).

▼ Choice b is incorrect as an exception:

★ SIADH (Syndrome of Inappropriate Antidiuretic Hormone secretion) is another classic endocrine paraneoplastic syndrome caused by ectopic ADH secretion, predominantly associated with small cell lung carcinoma.

▼ Choice d is incorrect as an exception:

👉 Carcinoid syndrome is a paraneoplastic syndrome that can occur with bronchial carcinoid tumors, presenting with vasomotor disturbances (flushing), diarrhea, and bronchospasm due to vasoactive amines (serotonin) released into the systemic circulation.

▼ Choice e is incorrect as an exception: Hypertrophic pulmonary osteoarthropathy (HPOA), also known as Marie-Bamberger syndrome, is characterized by periostitis and digital clubbing, and is primarily associated with lung adenocarcinomas.

✱ Core Pathology Note:

✂ Paraneoplastic Syndromes in Lung Cancer

📌 Small Cell Lung Carcinoma (SCLC):

★ SIADH (ADH secretion) and ★ Cushing's Syndrome (ACTH secretion).

📌 Squamous Cell Carcinoma (SCC):

★ Hypercalcemia due to ectopic production of PTHrP.

📌 Adenocarcinoma:

★ HPOA / clubbing and hematological syndromes (migratory thrombophlebitis).

Question 38:

▼ Restrictive lung disease

A. is almost always associated with wheezing

B. leads to increased oxygen diffusing capacity

C. is associated with reduced compliance

D. at end stage is referred to as jelly bean lung

E. is rarely associated with environmental diseases

✔ Answer is C (is associated with reduced compliance)

Explanation of Choices:

▼ Choice c is correct:

★ Restrictive lung diseases are characterized by diffuse, chronic involvement of the pulmonary connective tissue (interstitium).

★ The accumulation of inflammatory cells and collagen (fibrosis) stiffens the lung parenchyma, which severely reduces lung compliance (i.e., more pressure is required to expand the lungs during inspiration).

▼ Choice a is incorrect: Wheezing is a classic high-pitched expiratory sound caused by airway narrowing and is the clinical hallmark of obstructive airway diseases (like asthma), not restrictive lung disease.

▼ Choice b is incorrect:

★ Interstitial fibrosis thickens the alveolar-capillary barrier, which decreases (rather than increases) oxygen diffusing capacity (DLCO).

▼ Choice d is incorrect:

★ In advanced, end-stage restrictive lung disease, severe scarring and cystic destruction of alveolar structures create a characteristic gross appearance known as honeycomb lung (not "jelly bean lung").

▼ Choice e is incorrect:

★ Many restrictive lung diseases are directly linked to environmental diseases, specifically pneumoconioses (caused by inhaled mineral dusts like asbestosis, silicosis, and coal workers' pneumoconiosis) and hypersensitivity pneumonitis.

★ Core Pathology Note: Restrictive Lung Disease Hallmarks

💀 Pathology:

👉 Reduced lung expansion with decreased total lung capacity (TLC).

★ Key Parameters:

▼ Decreased compliance (stiff lungs) and decreased diffusing capacity.

👤 End Stage:

🎯 Retraction of scars produces a gross cobblestone pleural surface and a microscopic honeycomb pattern.

Question 39:

▼ Defense mechanisms against pulmonary infection include

A. Mucous clearance by ciliated epithelium at the front of the nasal cavity

B. Alveolar macrophage clearance of bacteria deposited in the alveoli

C. Mucociliary action of alveolar epithelium

D. Vomiting reflex

E. All of the above

✅ Answer is B (Alveolar macrophage clearance of bacteria deposited in the alveoli)

★ Explanation of Choices:

✂ Choice b is correct:

★ The alveoli (the site of gas exchange) lack a mucus lining or ciliated cells. Therefore, the primary and most vital defense mechanism against bacteria or particulates that reach the terminal airspaces is phagocytosis and clearance by alveolar macrophages.

▼ Choice a is incorrect:

★ The very front of the nasal cavity (the nasal vestibule) is lined by stratified squamous epithelium containing coarse hairs (vibrissae) for filtering, not ciliated respiratory epithelium. The ciliated epithelium begins further back in the respiratory region of the nasal cavity.

▼ Choice c is incorrect:

☞ Alveolar epithelium consists of Type I and Type II pneumocytes, which are not ciliated. There is no mucociliary action in the alveoli; the "mucociliary escalator" stops at the terminal bronchioles.

▼ Choice d is incorrect:

★ The vomiting reflex is not a defense mechanism against pulmonary infections. In fact, if protective reflexes are compromised (e.g., in a coma or under anesthesia), vomiting can lead to the dangerous aspiration of gastric contents, triggering a lung abscess. The actual protective reflexes of the airways are the cough reflex and epiglottic reflex.

Choice e is incorrect because a, c, and d are false.

★ Core Pathology Note: Pulmonary Clearance Mechanisms

▼ Conducting Airways: Cleared by the mucociliary apparatus (cilia beat mucus upward to be swallowed or expectorated).

▼ Alveolar Space: Cleared exclusively by alveolar macrophages via phagocytosis.

🚬 Impairment: Cigarette smoke damages ciliary function. Alcohol and tobacco smoke impair alveolar macrophage phagocytic action, predisposing to pneumonia.

Question 40:

▼ Chronic bronchitis major morphological change involves

A. WBC infiltration

C. Decreased goblet cell number

C. Smooth muscle hypertrophy

D. Increased Reid index

✅ Answer is D (Increased Reid index)

Explanation of Choices:

▼ Choice d is correct:

★ The diagnostic, defining histological hallmark of chronic bronchitis in the large airways is the hypertrophy and hyperplasia of the submucosal mucus-secreting glands. The magnitude of this enlargement is assessed by the Reid index (the ratio of the thickness of the submucosal gland layer to the thickness of the bronchial wall between the epithelium and cartilage). While the normal Reid index is approximately 0.4, it is characteristically increased in chronic bronchitis.

▼ Choice a is incorrect: While an inflammatory infiltrate of white blood cells (CD8+ T cells, macrophages, and neutrophils) is present, chronic inflammatory infiltration is a common feature of many airway diseases and is not the major, specific diagnostic morphological parameter like the Reid Index.

▼ Choice b is incorrect: Chronic bronchitis is characterized by an increased number (hyperplasia/metaplasia) of mucin-secreting goblet cells in the surface epithelium, especially in smaller airways, to produce excess mucus.

▼ Choice c is incorrect: Smooth muscle cell hypertrophy and hyperreactivity is the classic, defining histological feature of bronchial asthma, not chronic bronchitis.